

# Rukhshan Haroon

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## EDUCATION

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**Tufts University**, Greater Boston, MA, USA.

*Ph.D. in Computer Science*, Sept. 2022 - Sept. 2026 (expected)

Specialization: Accessibility, Human-Computer Interaction (HCI), Artificial Intelligence

**Lahore University of Management Sciences (LUMS)**, Lahore, Pakistan.

*B.Sc. in Computer Science*, Sept. 2018 - May 2022

## SKILLS

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**Research:** Semi-Structured Interviews, Think-Aloud Method, User Study & Experimental Design, Quantitative Methods, Survey Design, AI Bias & Performance Evaluation, Large-Scale System Design.

**Technical:** Agentic AI, Full-Stack Development, Python, PyTorch, C/C++, Unity, MongoDB, Git.

**Selected Coursework:** Assistive Design (Mechanical Eng.), Generative AI for Social Impact, HCI for Disability, Deep Learning, Statistics, Advanced Programming, Data Science, Data Structures.

## WORK EXPERIENCE

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**Research Lead (AI, Accessibility, and HCI)**—Tufts University, Boston, MA. *Sept 2022 - present.*

- Designed and prototyped an LLM-based: (1) training platform that uses conversational simulations to help non-autistic users improve cross-neurotype conversational skills; (2) writing assistant to support autistic users with self-expression and interpretation of linguistic nuances.
- Designed and led user studies with **30+** autistic users to evaluate the aforementioned systems.
- Conducted an empirical study of LLMs on tasks involving evaluation of cross-neurotype communication differences; contributed a comprehensive LLM evaluation framework and dataset.

**Researcher (AI, Systems, and HCI)**—Tufts University, Boston, MA. *Jan - Dec 2024.*

- Designed a cost-saving proxy for LLM routing, reducing costs by up to **50%** without quality loss.
- Deployed the proxy in **3 countries** over **12+ months** supporting **85k+ user-requests**.

**Research Lead (Security, Privacy, and HCI)**—LUMS, Lahore, Pakistan. *May 2020 - Sept 2022.*

- Led mixed-methods research in Pakistan to study: (1) gender disparities among healthcare workers in COVID-19; (2) how socio-cultural factors shaped pandemic-related misinformation spread.
- Implemented a benchmarking framework to compare CVEs, size, correctness, and build time of software debloating tools; built multi-tool pipelines achieving **70%** size and **40%** CVE reductions.

## SELECTED RESEARCH

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ASSETS is the flagship ACM SIGACCESS conference and the premier venue on accessible HCI.

**Rukhshan Haroon**, Kyle Wigdor, Katie Yang, Nicole Toumanios, Eileen Crehan, and Fahad Dogar. 2025. *NeuroBridge: Using Generative AI to Bridge Cross-neurotype Communication Differences through Neurotypical Perspective-taking*. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*. 🏆 **Best Student Paper** [PDF](#)

**Rukhshan Haroon** and Fahad Dogar. 2024. *TwIPS: A Large Language Model Powered Texting Application to Simplify Conversational Nuances for Autistic Users*. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24)*. [PDF](#)

**Rukhshan Haroon**, Haisum Haroon, Wren Kritzer, Jaylin Herskovitz, and Fahad Dogar. 2026. *Bridging or Widening the Gap? An Empirical Study on LLM Bias Against Autistic Communication Styles*. Under Submission at ACM ASSETS 2026. [PDF](#)

**Rukhshan Haroon**. 2026. *Towards AI-Mediated Bidirectional Communication Support for Cross-Neurotype Interactions*. SIGACCESS Accessibility and Computing. [PDF](#)

**Rukhshan Haroon**, Ayesha Naeem, Priya Sajjad, and Zartash Uzmi. 2024. *On the Frontline During the COVID-19 Pandemic: Gender Inequality and Experiences of Healthcare Workers in Pakistan*. *ACM Journal on Computing and Sustainable Societies*. [PDF](#)

## SERVICE AND VOLUNTEERING

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**Reviewer**, ACM CHI, ACM IUI, ACM TACCESS, and ACM UIST 2025

## HONORS AND AWARDS

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**Best Student Paper**, ACM ASSETS, [Press](#) 2025

**NSF Travel Award**, ACM ASSETS Doctoral Consortium 2025

**Augmented Reality Hackathon Winner**, Harvard University, [Blog](#) 2023

**Award of Academic Distinction**, LUMS 2020–2022

**Students as Co-Researchers Grant**, LUMS 2021